

JAIR FERNANDO FAJARDO-ROJAS

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Professional Appointments

Colorado School of Mines

Golden, CO

Postdoctoral Fellow, Department of Physics
Institute for Data Driven Dynamical Design – ID4
Advisor: Prof. Eric Toberer

Sept. 2025 – Present

Education

Colorado School of Mines

Golden, CO

Ph.D. in Materials Science, Department of Chemical & Biological Engineering
Advisor: Prof. Diego Gómez-Gualdrón

2021 – 2025

Thesis: “Data-Driven Development of Metal-Organic Frameworks: Synthesizability Insights and Thermodynamic Stability Predictions”

Universidad de los Andes

Bogotá, Colombia

Ph.D. in Engineering, Department of Chemical Engineering

2017 – 2021

Advisors: Prof. Diego Pradilla, Prof. Oscar Alvarez

Thesis: “Interfacial Behavior of Natural and Model Asphaltenes: A Multi-Scale Approach”

M.S. Chemical Engineering, Department of Chemical Engineering

2019 – 2021

Universidad Industrial de Santander

Bucaramanga, Colombia

M.S. Hydrocarbon Engineering, Department of Petroleum Engineering

2013 – 2016

B.S. Chemical Engineering, Department of Chemical Engineering

2008 – 2013

Honors & Awards

2024	Excellence Award Institute of Data-Driven Dynamics Design (ID4) – NSF For academic & community contributions
2024	ID4 Travel Award To attend “AI≡Science: Strengthening the Bond Between the Sciences and Artificial Intelligence” Workshop at UC, Berkeley
2018	Fulbright Scholarship – Colombian Doctoral Student Fulbright Colombia Travel & Stipend
2017 – 2021	Excellence Scholarship for Doctoral Studies Universidad de los Andes, Colombia
2013 – 2016	Ecopetrol Sponsorship for Master Studies Universidad Industrial de Santander, Colombia
2013	Distinguished Student Chemical Engineering Universidad Industrial de Santander, Colombia

Research Highlights

- Revealed functionalization role in MOF thermodynamic stability via high-throughput simulation and data-driven analysis.
- Developed a protocol to accelerate free energy of solvation calculation via molecular simulation in nanoporous materials.
- Designed data-efficient ML strategies to predict MOF free energy and adsorption properties.
- Developed material representations enabling ML predictions of porous materials adsorption properties and stability.
- Experimental identification of limitations for model asphaltene reproducibility of interfacial behavior of natural asphaltene.
- Macroscopic–microscopic relationships of the rheological behavior of asphaltene dynamical aggregation.

Proposals & Computational Resources

PI, Generative design of thermodynamically stable metal-organic frameworks via symmetry-aware diffusion models. Computational award: 200,000 core-hour credits – NSF-ACCESS granted (2026)

Contributor, Transferable machine learning (ML) potentials to enable ML-based screening of adsorbents for separations involving chemisorption. Award amount: \$320,117 – NSF (2025) | **PI**: Prof. Diego Gómez-Gualdrón.

Publications

* Equal contribution | [Google Scholar](#)

- (15) D. A. Gómez-Gualdrón, T.G. de Vilas, K. Ardila, **F. Fajardo-Rojas**, A. Pak. “Machine Learning to Design Metal–Organic Frameworks: Progress and Challenges from a Data Efficiency Perspective.” *Mater. Horiz.*, **2026**, *13*, 1694–1715. 10.1039/D5MH01467K
- (14) **F. Fajardo-Rojas**, R. Anderson, K. Ardila, A. Pak, D.A. Gómez-Gualdrón. “Interactions of Common Synthesis Solvents with MOFs Studied via Free Energies of Solvation: Implications on Stability and Polymorph Selection.” *Chem. Mater.* **2026**, *38*, 2, 607–618. 10.1021/acs.chemmater.5c01410
- (13) A. N. Rubungo*, **F. Fajardo-Rojas***, D. A. Gómez-Gualdrón, A.B. Dieng. “Highly Accurate and Fast Prediction of MOF Free Energy Via Machine Learning.” *J. Am. Chem. Soc.* **2025**, *147*, 52, 48035–48045. 10.1021/jacs.5c13960
- (12) **F. Fajardo-Rojas**, R. Anderson, M. Li, R. Chang, D.A. Gómez-Gualdrón. “Data-Driven Insights on the Impact of Functionalization on Metal–Organic Framework Free Energies.” *Chem. Mater.* **2025**, *37*, 15, 5502–5514. 10.1021/acs.chemmater.5c00129
- (11) S. Li, Fernando Fajardo-Rojas, D. A. Gómez-Gualdrón, R. Chang, M. Li. “AlloyLens: A Visual Analytics Tool for High-Throughput Alloy Screening and Inverse Design.” *Arxiv preprint*, **2025**. 10.48550/arXiv.2511.02133
- (10) X. Zhao, **F. Fajardo-Rojas**, J. Furst, K. Ardila, K. Langlois, Y. An, X. Hu, F. Uribe-Romo, D.A. Gómez-Gualdrón, J. Greenberg. “Expert-Guided LLM Approach for Sequence-Aware Extraction of MOF Synthesis.” *ChemRxiv preprint*, **2025**. 10.26434/chemrxiv-2025-x90hc
- (9) T.W. Liu, **F. Fajardo-Rojas**, S. Addish, E. Martinez, D.A. Gómez-Gualdrón. “MOFs to Enhance Green NH₃ Synthesis in Plasma Reactors: Hierarchical Computational Screening Enhanced by Iterative Machine Learning.” *ACS Appl. Mater. Interfaces* **2024**, *16*, 49, 68506–68519. 10.1021/acsami.4c11396
- (8) E. Osaro, **F. Fajardo-Rojas**, G.M. Cooper, D.A. Gómez-Gualdrón, Y.J. Colón. “Active Learning of Alchemical Adsorption Simulations: Towards a Universal Adsorption Model.” *Chem. Sci.*, **2024**, *15*, 17671–17684. 10.1039/D4SC02156H
- (7) S.S. Rajasree, J. Yu, **F. Fajardo-Rojas**, H.C. Fry, R. Anderson, X. Li, W. Xu, J. Duan, S. Goswami, K. Maindan, D.A. Gómez-Gualdrón, P. Deria. “Framework-Topology-Controlled Singlet Fission in Metal–Organic Frameworks.” *J. Am. Chem. Soc.* **2023**, *145*, 32, 17678–17688. 10.1021/jacs.3c03918
- (6) J.V. Pérez-Bejarano, **F. Fajardo-Rojas**, O. Alvarez, J.C. Burgos, L.H. Reyes, D. Pradilla. “Novel Biosurfactants: Rationally Designed Surface-Active Peptides and In-Silico Evaluation at the Decane-Water Interface.” *Process Biochem.*, **2023**, *125*, 84–95. 10.1016/j.procbio.2022.11.012
- (5) M. Vyas, **F. Fajardo-Rojas**, D.A. Gómez-Gualdrón, S. Kwon. “Theoretical Assessments of Pd–PdO Phase Transformation and Its Impacts on H₂O₂ Synthesis and Decomposition Pathways.” *Catal. Sci. Technol.*, **2023**, *13*, 3828–3848. 10.1039/D3CY00404J
- (4) Y. An, J. Greenberg, X. Hu, A. Kalinowski, X. Fang, X. Zhao, S. McClellan, F. Uribe-Romo, K. Langlois, J. Furst, D. A. Gómez-Gualdrón, **F. Fajardo-Rojas**, K. Ardila, S. K. Saikin, C. A. Harper, R. Daniel. “Exploring Pre-Trained Language Models to Build Knowledge Graph for Metal–Organic Frameworks (MOFs).” *IEEE International Conference on Big Data*, **2022**. 10.1109/BigData55660.2022.10020568
- (3) Y. An, J. Greenberg, X. Zhao, X. Hu, S. McClellan, A. Kalinowski, F. Uribe-Romo, K. Langlois, J. Furst, D. A. Gómez-Gualdrón, **F. Fajardo-Rojas**, K. Ardila. “Building Open Knowledge Graph for Metal–Organic Frameworks (MOF–KG): Challenges and Case Studies.” *Arxiv preprint*, **2022**. 10.48550/arXiv.2207.04502
- (2) **F. Fajardo-Rojas**, O. Alvarez, J.R. Samaniuk, D. Pradilla. “Deviation from Equilibrium Thermodynamics of an Asphaltene Model Compound During Compression–Expansion Experiments at Fluid–Fluid Interfaces.” *Langmuir* **2021**, *37*, 5, 1799–1810. 10.1021/acs.langmuir.0c03151
- (1) **F. Fajardo-Rojas**, D. Pradilla, O. Alvarez, J.R. Samaniuk. “Probing Interfacial Structure and Dynamics of Model and Natural Asphaltenes at Fluid–Fluid Interfaces.” *Langmuir* **2020**, *36*, 27, 7965–7979. 10.1021/acs.langmuir.0c01320

Under Review

T. G. de Vilas*, **F. Fajardo-Rojas***, O. Mansurov, R. Devaisher, E. Toberer, D. A. Gómez-Gualdrón. “Two-Dimensional Interaction Parameter Histograms as a Simple and Versatile Nanoporous Material Representation for Machine Learning Prediction of Adsorption Properties.” *Submitted to Molecular Systems Design & Engineering*, **2026**.

Research Experience

Colorado School of Mines	Golden, CO
Postdoctoral Fellow , Department of Physics	2025 – present
Building AI-enabled, multiscale computational workflows for predictive materials design, emphasizing integration between simulation, data-driven learning, and experimentation.	
Research Assistant , Department of Chemical and Biological Engineering	2021 – 2025
Developed computational frameworks to accelerate data-driven analysis and machine learning-based prediction of adsorption and thermodynamic stability in porous materials.	
Universidad de los Andes	Bogotá, Colombia
Research Assistant , Department of Chemical Engineering	2017 – 2021
Conducted experiments to elucidate multi-scale structure–property relationships governing the behavior of surface-active molecules at fluid–fluid interfaces for colloidal systems design.	
Project Specialist , Department of Chemical Engineering & DOW Chemical Colombia	Summer 2021
Performance assessment of surfactant formulations in destabilizing crude oil–water emulsions.	
Universidad Industrial de Santander	Bucaramanga, Colombia
Researcher , Department of Petroleum Engineering	2015 – 2017
Evaluated combustion reaction kinetics under oil reservoir conditions to assess how oil composition and reservoir heterogeneity affect in-situ combustion performance for enhanced oil recovery.	
Enhanced Oil Recovery Engineer , Department of Petroleum Engineering & ECOPETROL	2016
Technical supervision of enhanced oil recovery (EOR) research and development in Colombia's petroleum fields.	

Teaching Experience

Colorado School of Mines, <i>Golden, CO</i>	Spring 2026
Guest Lecturer , Materials and Computation for Clean Energy (ENGY/MEGN 598)	
Universidad de los Andes, <i>Bogotá, Colombia</i>	2017 – 2021
Teaching Assistant , Reaction Kinetics (Dept. Chemical Eng., Spring 2017 – Spring 2018) Introduction to ChE, Thermodynamics (Dept. Chemical Eng., Fall 2019 – Spring 2021)	
Universidad Industrial de Santander, <i>Bucaramanga, Colombia</i>	2014 – 2022
Guest Lecturer , Enhanced Oil Recovery – Miscible Methods (Dept. Petroleum Eng., Fall 2022)	
Instructor , Thermodynamics, Transport Phenomena (Dept. Petroleum Eng., Spring 2014 – Fall 2016)	

Mentoring Experience

Graduate Students

In Mines

Colton Gerber	Ph.D. student, Mines Mat. Sci.	2025 – present
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Before Mines

Diego Ayala	M.S. student, Uniandes ChemE	Current: Analyst, HINICIO	2021 – 2023
Johana Pérez	M.S. student, Uniandes ChemE	Current: R+D Scientist, Danec	2019 – 2021

Undergraduate Students

In Mines

Omar Mansurov	CBE 26' GRFP (Committed to Northwestern for Ph.D.)	2025 – present
Diego Hernandez (REU)	Current: Miami Dade College, Chemistry	Summer 2025
Ruby Devaisher (REU)	Current: UIUC, Physics	Summer 2025
Luisa Ruiz (REU)	Current: Penn State, Ph.D. student Mat. Sci. & Eng.	Summer 2024

Jack Canonico (SURF)	QBE 26'	Summer 2024
Vashti Trujillo (REU)	Current: Colorado State U.– Pueblo, Mechatronics	Summer 2023
Enrique Martinez	CBE 24', Current: Chevron	2023
Sumaya Adish (REU)	Current: UNC–Chapel Hill, Ph.D. student Bio. Sci.	Summer 2022
Dale Baum	CBE 24'	2022
Candan Erdemir	CBE 23'	2022

Before Mines

Cesar Bucheli	Uniandes, ChemE	Current: Brown U., Ph.D. student Geoscience	2020
Jadier Aristizabal	UIS, Petroleum Eng		2017
Daniela Mojica	UIS, Petroleum Eng.		2017
Andres Vargas	UIS, Petroleum Eng.	Current: PSD Manager, Schlumberger	2016
Jenifer Fierro	UIS, Petroleum Eng.	Current: Data Technician, Phoenix Technology Service	2016
Lizet Rojas	UIS, Petroleum Eng.		2016
Clara Mendoza	UIS, ChemE	Current: Senior Process Specialist, Evac	2014

Presentations

Invited Presentations

2026 Jun.	Accelerating Quantum Materials Discovery. Colorado School of Mines. Golden, Colorado.
2025 Oct.	NSF Institute for Data-Driven Dynamics Design (ID4). Colorado School of Mines, Golden, Colorado.

Contributed Presentations

2025 Nov.	AIChE Annual Meeting, Boston, Massachusetts (Oral)
2025 Nov.	AIChE Annual Meeting, Boston, Massachusetts (Poster)
2024 Nov.	AIChE Annual Meeting, San Diego, California (Oral)
2024 Nov.	AIChE Annual Meeting, San Diego, California (Poster)
2024 Aug.	ACS Fall Meeting, Denver, Colorado (Oral)
2023 Nov.	AIChE Annual Meeting, Orlando, Florida (Oral)
2019 Nov.	AIChE Annual Meeting, Orlando, Florida (Oral)

Academic & Professional Service

Conference Organization and Service

Poster Judge , AIChE, UG Poster Competition	2025
Scientific Committee , Symposium on Hydrocarbon Research, Universidad Industrial de Santander, Colombia.	2024 – present

Journal Reviewer: Nature Communications, Journal of Alloys and Compounds, Scientific Reports.

Graduate Thesis Committee Member/Examiner

Sebastian Solarte, Ph.D. (Advisors: Oscar Alvarez & Sebastien Simon, Universidad de los Andes, Colombia)	2026 – present
Sergio Castellanos, M.S. (Advisor: Adan León, Universidad Industrial de Santander, Colombia)	2024
Mishel Porras, M.S. (Advisor: Diego Pradilla, Universidad de los Andes, Colombia)	2023
Bayron Torres, M.S. (Advisor: Emiliano Ariza, Universidad Industrial de Santander, Colombia)	2023
Jorge Diaz, M.S. (Advisor: Zuly Calderon, Universidad Industrial de Santander, Colombia)	2022

Outreach and Leadership Service

Selection committee , Fulbright Commission Colombia	2022 – present
Interdisciplinary Programs Liaison , Chemical Engineering Graduate Association (CEGA), Mines	2024
Materials Science Representative , Graduate Students Government (GSG), Mines	2022
President , Society of Petroleum Engineers, Uniandes Student Chapter	2017 – 2019